

KOMATSUJIMA AB, Japan

WMO: 478840

Lat: 34.005N

Lon: 134.626E

Elev: 19

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.9	35.6	19.3	14.5	38.1	21.3	16.0	39.4	32.3	44.9	30.1	44.7	13.5	300	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.2	91.1	78.8	89.4	78.0	87.7	78.0	80.1	87.4	79.4	86.4	78.7	85.3	12.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.7	144.0	84.4	77.2	141.5	83.8	76.7	139.2	83.3	43.8	88.1	42.9	86.8	42.1	85.4	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.0	26.7	23.9	30.3	93.9	2.6	2.3	28.5	95.5	26.9	96.8	25.5	98.1	23.6	99.7
			26.4	81.8	1.7	1.3	25.2	82.8	24.2	83.6	23.2	84.3	22.0	85.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	62.5	43.9	44.5	49.3	58.2	66.6	73.3	80.1	82.5	76.9	67.6	57.6	48.4	(6)
(7)		DBStd	14.07	3.82	4.51	4.96	5.24	3.72	3.69	3.54	2.68	4.42	4.39	5.09	4.73	(7)
(8)		HDD50	522	192	163	74	4	0	0	0	0	0	3	86	(8)	
(9)		HDD65	2710	653	573	486	208	26	0	0	0	22	228	514	(9)	
(10)		CDD50	5094	3	10	53	252	516	698	933	1006	808	547	230	38	(10)
(11)		CDD65	1806	0	0	0	6	77	248	468	541	358	104	4	0	(11)
(12)		CDH74	14292	0	0	0	11	158	1049	4305	5904	2645	219	1	0	(12)
(13)		CDH80	4241	0	0	0	0	11	141	1330	2125	621	13	0	0	(13)
(14)	Wind	WSAvg	12.0	14.2	13.8	13.5	12.5	11.4	9.6	9.8	10.3	11.2	11.7	12.3	13.7	(14)
(15)	Precipitation	PrecAvg	65.2	1.9	2.6	3.9	4.3	5.9	8.0	6.9	7.4	10.7	6.8	4.1	2.7	(15)
(16)		PrecMax	98.2	4.4	5.8	7.0	8.9	17.0	15.0	15.9	37.5	33.4	18.5	11.1	9.8	(16)
(17)		PrecMin	39.1	0.3	0.6	1.8	1.6	1.8	2.2	2.5	0.9	2.7	0.7	0.6	0.2	(17)
(18)		PrecStd	16.1	1.1	1.4	1.5	1.9	3.7	3.0	3.3	7.7	6.9	4.4	2.8	2.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.8	63.6	67.6	75.3	80.9	87.8	93.1	93.4	90.0	82.0	72.0	63.9	(19)
(20)		0.4%	MCWB	54.5	58.6	59.5	63.0	66.6	77.2	78.3	79.6	77.8	73.2	63.8	57.6	(20)
(21)		2%	DB	54.2	57.7	64.2	71.7	77.4	83.8	90.0	91.5	87.8	78.6	69.8	60.0	(21)
(22)		2%	MCWB	48.0	52.1	57.5	62.7	65.9	74.4	78.1	79.2	77.2	70.4	62.9	53.3	(22)
(23)		5%	DB	52.1	54.0	60.8	69.4	75.4	80.9	88.2	89.8	85.8	76.6	67.9	57.5	(23)
(24)		5%	MCWB	46.1	48.2	54.5	60.8	65.3	72.6	78.2	78.2	75.8	68.5	61.5	50.8	(24)
(25)		10%	DB	50.2	51.8	58.4	66.5	73.4	79.1	86.3	88.1	83.8	74.4	65.9	55.6	(25)
(26)		10%	MCWB	44.1	46.2	52.3	58.9	64.7	71.9	77.7	78.1	75.0	66.6	59.7	49.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.9	59.3	62.0	67.9	72.9	78.5	80.7	81.6	80.0	75.0	67.4	59.7	(27)
(28)		0.4%	MCDWB	58.6	62.8	64.9	71.8	76.3	85.1	88.4	89.6	86.2	79.7	69.7	62.0	(28)
(29)		2%	WB	50.6	53.3	58.8	65.0	69.9	76.5	79.7	80.5	78.8	72.2	64.7	54.8	(29)
(30)		2%	MCDWB	53.0	56.9	62.8	69.3	73.4	81.1	87.0	88.1	84.8	76.7	68.2	58.9	(30)
(31)		5%	WB	47.3	50.1	55.7	62.5	67.8	75.3	79.0	79.8	77.7	69.9	62.3	52.1	(31)
(32)		5%	MCDWB	51.3	52.8	59.9	67.3	72.1	79.2	85.9	87.1	83.3	74.2	66.5	56.0	(32)
(33)		10%	WB	44.6	47.3	52.9	60.3	66.2	74.0	78.2	79.1	76.5	67.9	60.1	50.2	(33)
(34)		10%	MCDWB	49.2	50.9	57.2	65.3	71.7	77.7	84.7	86.0	81.9	73.0	64.7	54.5	(34)
(35)	Mean Daily Temperature Range		MDBR	9.7	10.6	12.2	12.9	12.2	10.2	10.3	11.2	9.7	9.6	10.0	9.8	(35)
(36)		5% DB	MCDBR	12.9	14.2	16.4	16.4	15.7	12.6	13.1	13.3	11.8	11.4	12.0	12.3	(36)
(37)			MCWBR	10.0	11.2	12.8	10.0	8.4	6.1	5.5	5.9	5.7	6.8	8.5	9.0	(37)
(38)			MCDBR	12.5	13.1	15.1	13.5	12.4	10.1	11.8	12.0	10.3	9.6	11.3	11.6	(38)
(39)		5% WB	MCDWBR	11.2	11.8	13.7	10.1	8.5	5.7	5.6	5.9	5.7	7.1	9.5	10.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.399	0.447	0.508	0.540	0.547	0.562	0.561	0.531	0.496	0.459	0.423	0.393	(40)	
(41)		taud	2.175	2.056	1.911	1.883	1.938	1.976	2.025	2.098	2.167	2.187	2.193	2.208	(41)	
(42)		Ebn at Noon	248	248	243	242	242	238	237	242	245	243	239	242	(42)	
(43)		Edh at Noon	37	47	58	63	60	58	55	50	45	41	36	34	(43)	
(44)	All-Sky Solar Radiation	RadAvg	820	1005	1309	1545	1646	1453	1627	1683	1308	1066	834	736	(44)	
(45)		RadStd	59	77	92	154	165	151	195	150	113	110	73	47	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page